

Prior-Art Reconnaissance — The Quantum Computing Landscape

Dossier QC-Landscape · Phase 0 working map (pre-release draft)

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STATUS: Phase 0 prior-art reconnaissance — a working map, not laddered claims. Nothing here is asserted as a verified result of this dossier. Every entry is tagged by evidence tier (ESTABLISHED / ADJACENT / OPEN). Demonstrated, peer-reviewed results are stated plainly with sources; vendor roadmaps, announcements, and contested claims are flagged as such; comparative and predictive judgements are held in the OPEN bucket as forecasts. These rows still need a per-claim human reality-check before any of them enters the formal claim ledger, `verify_numbers.py`, or the manuscript. *The one rule: every label is true.*

How to read this map

One distinction governs everything below, because it is the single most common way this topic gets misreported:

- **Physical vs logical qubits.** A raw qubit count is not an error-corrected (logical) qubit count. We state which is which everywhere.
- **Demonstrated vs roadmap.** A measured, published result is a different kind of object from a company's projection of where it will be in 2029. Both appear here; only the first is ESTABLISHED.
- **Peer-reviewed vs interested-party.** A Nature paper and a press release are not the same evidence tier. Vendor and aggregator claims are flagged.

Evidence tiers used here:

- **ESTABLISHED** — demonstrated on hardware, with a peer-reviewed or otherwise primary source. Asserted plainly.
 - **ADJACENT** — announced, roadmapped, or contested; an interested party's claim not yet independently reproduced, or a result the community disputes.
 - **OPEN** — a comparative or predictive judgement (probability of success, "who wins," timelines). Held as a forecast with a falsifiable signpost; never asserted as settled.
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The landscape, avenue by avenue

Superconducting (Google, IBM, Rigetti, IQM)

- **Best demonstrated:** Google's Willow processor (105 physical qubits) showed *below-threshold* error correction — the logical error rate falls as the code distance grows (d=3 to d=5 to d=7),

reaching roughly 0.14% logical error per cycle at $d=7$ with real-time decoding (peer-reviewed, late 2024). IBM's Condor reached 1,121 physical qubits in 2023 before IBM pivoted to smaller, higher-quality, modular chips (Heron, Nighthawk).

- **Strength:** most mature at scale; fast gates; the clearest public access ladder and roadmap (IBM).
- **Weakness:** surface-code overhead (hundreds-to-thousands of physical qubits per logical qubit); cryogenic; chip-to-chip interconnect is the scaling frontier.
- **Maturity:** highest. The reference platform the others are measured against.

Trapped ion (Quantinuum, IonQ — IonQ now includes Oxford Ionics)

- **Best demonstrated:** Quantinuum's Helios — 98 physical barium ions at ~99.92% two-qubit fidelity across all pairs, yielding 48 error-corrected logical qubits at roughly a 2:1 physical-to-logical ratio (Nov 2025). Oxford Ionics / IonQ have reported the highest gate fidelities of any platform (two-qubit ~99.99%, single-qubit error around the 10^{-7} level).
- **Strength:** highest fidelities and longest coherence of any modality; the best (lowest) encoding ratio on the logical-qubit leaderboard.
- **Weakness:** slower gate speeds; laser/optics complexity has been the historic scaling bottleneck (the problem the microwave-gate and ions-on-chip approaches are trying to solve).
- **Maturity:** high on quality; scaling to large qubit numbers is the open question.

Neutral atom (QuEra, Pasqal, Atom Computing)

- **Best demonstrated:** QuEra reported 96 logical qubits from 448 physical atoms with below-threshold error suppression (peer-reviewed, Jan 2026) — the current logical-qubit count leader. Atom Computing crossed 1,000 physical atoms in 2023; Pasqal has operated ~1,000-atom systems.
- **Strength:** traps are cheap; atoms are identical by physics (no device-to-device variability); connectivity is reconfigurable in software rather than fixed by lithography.
- **Weakness:** a younger control/gate stack; gate speed and fidelity still maturing relative to ions.
- **Maturity:** rising fast; strongest recent logical-qubit results.

Silicon spin (Intel, Diraq, Silicon Quantum Computing, Quantum Motion, Equal1)

- **Best demonstrated:** Diraq + imec showed >99% two-qubit fidelity for silicon spin qubits built on a standard 300mm CMOS foundry line — and, notably, across multiple randomly selected dies, not one hero device (peer-reviewed, Sept 2025). Silicon Quantum Computing reported ~99.99% two-qubit fidelity (peer-reviewed, Dec 2025).
- **Strength:** smallest physical footprint; rides the existing semiconductor manufacturing base, giving (in principle) the cheapest path to millions of qubits.
- **Weakness:** lowest qubit counts today; no logical-qubit demonstration yet on this modality.
- **Maturity:** earliest on integration, but the manufacturing-scale argument is the strongest of any avenue if the counts come.

Photonic (PsiQuantum, Xanadu, Quandela)

- **Best demonstrated:** Xanadu's Aurora — a 12-qubit modular, networked, room-temperature photonic machine built from 35 chips and ~13 km of fiber (peer-reviewed, Jan 2025) — plus on-chip generation of error-resistant GKP states (peer-reviewed, June 2025).
- **Strength:** room-temperature operation; networking and modularity are native; rides silicon-photonics fabrication.
- **Weakness: optical loss** is the make-or-break bottleneck; many gates are inherently probabilistic, demanding heavy redundancy.
- **Maturity:** key building blocks shown; the loss problem stands between the blocks and a useful machine.

Topological (Microsoft) — CONTESTED

- **Claim:** Majorana 1 (Feb 2025), presented as 8 topological qubits and a hardware path toward a million-qubit chip.
- **Status — ADJACENT / contested:** the accompanying Nature paper's reviewers explicitly noted the results do not constitute evidence for Majorana zero modes, and independent physicists at the 2025 APS meeting were largely unconvinced the device demonstrates a topological qubit. A follow-up (Majorana 2) reports longer lifetimes but leaves the central physics question open.
- **If real:** intrinsic, hardware-level error protection — a potentially decisive scaling advantage. That payoff is exactly why it is worth pursuing, and exactly why the bar for "demonstrated" must stay high.
- **Maturity:** a genuine materials-science bet whose foundational claim is not yet accepted.

Annealing (D-Wave) — a distinct, non-gate avenue

- **Status:** D-Wave's Advantage2 annealers are in commercial optimization use today; annealing is not universal/gate-based computing. D-Wave has now announced a separate gate-model roadmap (initial small system in 2026; a target of 100 logical qubits by 2032) — that gate-model line is early and roadmap-stage.
- **Note:** keep annealing clearly separated from the gate-based fault-tolerance race; it answers a different question.

Sub-threads worth a flag (not top-level rows here)

- **Cat / bosonic qubits (Alice & Bob):** hardware-level bit-flip protection, with reported bit-flip lifetimes exceeding one hour — a distinct error-correction strategy that can sit under superconducting hardware.
- **NV-center / diamond, and Rydberg-specific variants:** real but treated as sub-threads pending your call on whether any deserves its own row.

The three buckets (summary)

ESTABLISHED (demonstrated; mostly peer-reviewed)

1. Below-threshold error correction now exists on real hardware in at least three modalities — superconducting (Google Willow), neutral-atom (QuEra), and trapped-ion (Quantinuum Helios). It is no longer only theory.
2. Trapped ions hold the fidelity lead; silicon spin has now matched ~99.99% two-qubit fidelity *and* shown it on an industrial foundry line across non-cherry-picked dies.
3. High-rate codes (color codes, qLDPC) have driven the physical-to-logical ratio down to ~2:1 on ion and atom hardware, versus hundreds-to-one for the surface code — a demonstrated overhead reduction.
4. Neutral-atom and superconducting platforms operate at 1,000+ physical qubits in real systems; photonics has shown a room-temperature modular networked prototype.

ADJACENT (announced / roadmap / contested — interested-party until reproduced)

1. Every "fault tolerance by year X" roadmap: IBM (large-scale FT ~2029), IonQ (an accelerated roadmap citing thousands of logical qubits by the late 2020s), Pasqal / QuEra / Atom logical-qubit ladders, D-Wave's gate-model 100-logical-by-2032. All are vendor projections.
2. Microsoft's topological qubit (Majorana 1 / 2): a real materials bet whose foundational claim the community disputes.
3. Single-vendor "quantum advantage" / utility benchmarks not yet independently reproduced.

OPEN (the dossier's core forecasts — to be posted as challenges)

1. Which modality reaches fault-tolerant, useful-scale QC first.
2. The probability of success per avenue/material.
3. The "most promising candidate" call.
4. The timeline to a cryptographically-relevant or industrially-useful machine.

Each OPEN item, when entered in the ledger, carries a forecast label, the explicit reasoning and a base rate or comparable where possible, and a falsifiable signpost — for example: a logical qubit sustained below a stated error rate over a stated number of cycles; a below-threshold surface-code patch at a stated scale; or a photonic system demonstrating loss below the fault-tolerance threshold.

Candidate OPEN forecasts to seed the ledger (draft — needs sign-off)

These are *not yet* ledger rows. They are the predictive claims this dossier will make, each to be labeled OPEN-UNVERIFIED with a signpost:

- **QC-F1** — "No single modality has yet secured the path to fault-tolerant, useful-scale QC; the leaders by distinct metric are trapped-ion (fidelity, encoding ratio), neutral-atom (logical-qubit

count), and superconducting (scale, maturity)." (Forecast; signpost: first sustained, reproduced logical-qubit advantage on a useful problem.)

- **QC-F2** — Per-avenue probability-of-success estimates (to be built with stated assumptions). (Forecast; signposts per avenue.)
- **QC-F3** — "Silicon spin has the strongest manufacturing-scale argument but the furthest to go on logical qubits." (Forecast; signpost: first logical-qubit demonstration on silicon spin at a stated fidelity.)
- **QC-F4** — "Microsoft's topological route remains unproven at the level of the qubit itself." (Currently ADJACENT/contested; signpost: independent reproduction of a topological qubit with accepted Majorana evidence.)

Reality-check questions for the author

1. **Aggregator-sourced numbers.** Several figures (some fidelity records, logical-qubit counts, roadmap dates) trace to industry-news aggregators that often restate vendor PR. The Nature-anchored ones (Willow, QuEra 96, Diraq/imec, Xanadu, SQC) are firmer. Any you know to be misreported?
2. **The probability call.** The safe framing is "no clear winner yet," but this dossier deliberately wants a probability-of-success call per avenue. Do you have a prior on which avenue you lean toward, so the strongest forecast is built there?
3. **Missing avenues.** Should cat/bosonic, NV-center/diamond, or any Rydberg variant be promoted to a top-level row?
4. **Materials axis.** You flagged materials specifically (silicon-28 enrichment; barium vs ytterbium ions; InAs/Al for topological; lithium-niobate vs silicon-nitride photonics). Develop "materials" as its own axis, or fold it into each modality?

Sources consulted (Phase 0; to be re-verified at citation-audit stage)

Peer-reviewed / primary:

- Silicon spin, 300mm foundry, >99% fidelity — *Nature* (2025): <https://www.nature.com/articles/s41586-025-09531-9>
- Diraq/imec milestone announcement — Diraq newsroom (Sept 2025): <https://diraq.com/newsdes/imec-technology-lights-the-path-to-utility-scale-for-diraq-s-quantum-chips>
- Xanadu Aurora (modular networked photonic) — Xanadu / *Nature* (Jan 2025): <https://www.xanadu.ai/press/xanadu-introduces-aurora-worlds-first-scalable-networked-and-modular-quantum-computer>
- Xanadu on-chip GKP states — The Quantum Insider (June 2025): <https://thequantuminsider.com/2025/06/05/xanadu-demonstrates-scalable-building-block-for-photonic-quantum-computers/>

- Microsoft topological-qubit skepticism — APS *Physics* (2025): <https://link.aps.org/doi/10.1103/Physics.18.68> ; *Science* (2025): <https://www.science.org/content/article/debate-erupts-around-microsoft-s-blockbuster-quantum-computing-claims>

Industry news / vendor (flagged — restate vendor or aggregator claims):

- Superconducting / Willow below-threshold, roadmaps — Network World (2025): <https://www.networkworld.com/article/4088709/top-quantum-breakthroughs-of-2025.html>
- Trapped-ion state of play (2026) — The Quantum Insider: <https://thequantuminsider.com/2026/06/12/trapped-ion-quantum-computing-companies-technology-and-where-it-stands-in-2026/>
- Trapped-ion fidelity / Helios detail — PostQuantum: <https://postquantum.com/quantum-modalities/trapped-ion-qubits/>
- Neutral-atom companies / QuEra 96 logical — Quantum Zeitgeist: <https://quantumzeitgeist.com/top-neutral-atom-quantum-computing-companies/>
- Pasqal roadmap — Pasqal newsroom (2026): <https://www.pasqal.com/newsroom/pasqal-releases-2025-roadmap/>
- IBM roadmap detail — PostQuantum: <https://postquantum.com/quantum-computing-companies/ibm/>
- Silicon-spin companies (2026) — Quantum Zeitgeist: <https://quantumzeitgeist.com/top-silicon-spin-quantum-computing-companies/>
- D-Wave gate-model roadmap (June 2026) — Business Wire: <https://www.businesswire.com/news/home/20260601444734/en/>
- Quantum error correction / logical-qubit leaderboard overview — Quantum Zeitgeist: <https://quantumzeitgeist.com/what-is-quantum-error-correction/>
- "What quantum computer to buy?" buyer-guide preprint (roadmap synthesis): <https://arxiv.org/pdf/2604.04761>

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